Answers learning task 15

2026 – 2027

Exercise 9.1: State whether each of the following situations is a correctly stated hypothesis testing problem and why.

(a) $H_0 : \mu = 25$, $H_1 : \mu \neq 25$
(b) $H_0 : \sigma > 10$, $H_1 : \sigma = 10$
(c) $H_0 : \overline{x} = 50$, $H_1 : \overline{x} \neq 50$
(d) $H_0 : p = 0.1$, $H_1 : p = 0.5$
(e) $H_0 : s = 20$, $H_1 : s < 20$

Answer

Notice that correctly stated hypotheses have to satisfy the following properties:

- H_0 and H_1 are statements about the value of a certain distribution parameter (e.g., the value of μ or σ in a normal distribution, the success probability p in a binomial distribution, ...).
- H_0 is of the form "=", while H_1 is of the form "$< / \neq / >$".

(a) Correct.

(b) Incorrect, but would have been correct if H_0 and H_1 were swapped.

(c) Incorrect, as H_0 and H_1 say something about the value of the sample statistic $\overline{x}$, not the population parameter μ.

(d) Incorrect, as the alternative hypothesis H_1 is not of the right form. It would have been a correctly stated hypothesis testing problem if $H_1 : p \neq 0.1$.

(e) Incorrect, as again H_0 and H_1 say something about the value of a sample statistic s, not the population parameter σ.

Exercise 9.5: A textile fiber manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kg with a standard deviation of 1.5 kg. The company wishes to test the hypothesis $H_0 : \mu = 12$ against $H_1 : \mu < 12$, using a random sample of four specimens.

(a) What is the type I error probability if the critical region is defined as $\bar{x} < 11.5$ kg?

(b) Find β for the case in which the true mean elongation is 11.25 kg.

(c) Find β for the case in which the true mean is 11.5 kg.

Answer

(a) Under the null hypothesis, we assume that the thread elongation $X \sim N(\mu = 12, \sigma = 1.5)$. With other words, the mean thread elongation is normally distributed with $\mu = 12$ en $\sigma = \frac{1.5}{\sqrt{4}} = 0.75$. If the critical region is defined as $\overline{x} < 11.5$, that

means the type I error probability α (the probability we reject H_0 while it is true) is equal to

$$\begin{aligned}\alpha &= P(\overline{X} < 11.5 | \mu = 12) \\ &= \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = 11.5, \mu = 12, \sigma = 0.75) \\ &\approx 0.2525.\end{aligned}$$

(b) We compute β (type II error probability) as the probability that H_0 will NOT be rejected, while H_1 is true (some given value of μ_1 satisfying H_1). Notice that $\mu_1 = 11.25$ satisfies $H_1 : \mu < 12$. We reject the null hypothesis if the sample mean $\overline{x} < 11.5$, or alternatively, we fail to reject H_0 if $\overline{x} \geq 11.5$:

$$\begin{aligned}\beta &= P(\overline{X} \geq 11.5 | \mu_1 = 11.25) \\ &= \text{normalcdf}(\text{lower} = 11.5, \text{upper} = 10^{99}, \mu = 11.25, \sigma = 0.75) \\ &\approx 0.3694.\end{aligned}$$

(c) Notice that this given value of $\mu_1 = 11.5$ is equal to the boundary value of the critical region. That means that under this true mean, the probability of a value inside the critical region is $50\,\%$, and also for the acceptance region. This means that we can directly tell that $\beta = 0.5$.

Exercise 9.7: In Exercise 9.5, find the boundary of the critical region if the type I error probability is

(a) $\alpha = 0.01$ and $n = 5$
(b) $\alpha = 0.05$ and $n = 5$
(c) $\alpha = 0.01$ and $n = 20$
(d) $\alpha = 0.05$ and $n = 20$

Answer

Notice that the hypothesis test of Exercise 9.5 can be formulated as $H_0 : \mu = 12$ versus $H_1 : \mu < 12$. That means that we are considering a left-sided hypothesis test.

In a hypothesis testing, we assume H_0 is true unless the evidence of the sample gives us significant reason to reject it. Under assumption of H_0, the mean thread elongation $\overline{X}$ is normally distributed with $\mu = 12$ and $\sigma = \frac{1.5}{\sqrt{n}}$. In a left sided test, that means that the critical region is of the form $(-\infty, g]$, where g is the α-th percentile of this normal distribution:

$$g = \text{InvNorm}(\text{area} = \alpha, \mu = 12, \sigma = \frac{1.5}{\sqrt{n}})$$

(a) Filling in $\alpha = 0.01$ and $n = 5$ gives $g \approx 10.4394$
(b) Filling in $\alpha = 0.05$ and $n = 5$ gives $g \approx 10.8966$
(c) Filling in $\alpha = 0.01$ and $n = 20$ gives $g \approx 11.2197$
(d) Filling in $\alpha = 0.05$ and $n = 20$ gives $g \approx 11.4483$

Exercise 9.9: In Exercise 9.5, calculate the p-value if the observed statistic is

(a) $\bar{x} = 11.25$ (b) $\bar{x} = 11.0$ (c) $\bar{x} = 11.75$

Answer

Notice that the hypothesis test of Exercise 9.5 can be formulated as $H_0 : \mu = 12$ versus $H_1 : \mu < 12$. That means that we are considering a left-sided hypothesis test.

In a hypothesis testing, we assume H_0 is true unless the evidence of the sample gives us significant reason to reject it. Under assumption of H_0, the mean thread elongation $\overline{X}$ is normally distributed with $\mu = 12$ and $\sigma = \frac{1.5}{\sqrt{n}}$. In a left-sided test, that means that the p-value (the probability that the test statistic $\overline{X}$ takes an even more extreme value than the observed value $\overline{x}$) is computed as $P(\overline{X} < \overline{x})$:

$$p = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = \overline{x}, \mu = 12, \sigma = \frac{1.5}{\sqrt{4}} = 0.75)$$

(a) Filling in $\overline{x} = 11.25$ gives $p \approx 0.1587$.

(b) Filling in $\overline{x} = 11.0$ gives $p \approx 0.0912$.

(c) Filling in $\overline{x} = 11.75$ gives $p \approx 0.3694$.

Exercise 9.15: A consumer products company is formulating a new shampoo and is interested in foam height (in millimeters). Foam height is approximately normally distributed and has a standard deviation of 25 mm. The company wishes to test $H_0 : \mu = 175$ mm versus $H_1 : \mu > 175$ mm, using the results of $n = 15$ samples.

(a) Find the type I error probability α if the critical region is $\overline{x} > 185$.

(b) What is the probability of type II error if the true mean foam height is 185 mm?

(c) Find β for the true mean of 195 mm.

Answer

(a) Under the null hypothesis, we assume that the foam height $X \sim N(\mu = 175, \sigma = 25)$. With other words, the mean foam height of 15 samples is normally distributed with $\mu = 175$ en $\sigma = \frac{25}{\sqrt{15}} \approx 6.4550$. If the critical region is defined as $\overline{x} > 185$, that means the type I error probability α (the probability we reject H_0 while it is true) is equal to

$$\alpha = P(\overline{X} > 185 | \mu = 175) = \text{normalcdf}(\text{lower} = 185, \text{upper} = 10^{99}, \mu = 175, \sigma = 6.4550) \approx 0.0607$$

(b) Notice that this given value of $\mu_1 = 185$ is equal to the boundary value of the critical region. That means that under this true mean, the probability of a value inside the critical region is $50\,\%$, and also for the acceptance region. This means that we can directly tell that $\beta = 0.5$.

(c) We compute β (type II error probability) as the probability that H_0 will NOT be rejected, while H_1 is true (some given value of μ_1 satisfying H_1). Notice that $\mu_1 = 195$ satisfies $H_1 : \mu > 175$. We reject the null hypothesis if the sample mean $\overline{x} > 185$, or alternatively, we fail to reject H_0 if $\overline{x} \leq 185$:

$$\beta = P(\overline{X} \leq 185 | \mu_1 = 195) = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = 185, \mu = 195, \sigma = 6.4550) \approx 0.06$$

Notice that this is the same answer as for (a), as $P(\overline{X} > 185 | \mu = 175) = P(\overline{X} < 185 | \mu = 175)$. This makes sense, as in both cases the mean μ is exactly 10 apart from the boundary value of the probability to be computed.

Exercise 9.17: In Exercise 9.15, find the boundary of the critical region if the type I error probability is

(a) $\alpha = 0.01$ and $n = 10$

Answer

Notice that the hypothesis test of Exercise 9.15 can be formulated as $H_0 : \mu = 175$ versus $H_1 : \mu > 175$. That means that we are considering a right-sided hypothesis test.

In a hypothesis testing, we assume H_0 is true unless the evidence of the sample gives us significant reason to reject it. Under assumption of H_0, the mean foam height $\overline{X}$ is normally distributed with $\mu = 175$ and $\sigma = \frac{25}{\sqrt{n}}$. In a right-sided test, that means that the critical region is of the form $[g, \infty)$, where g is the $(1 - \alpha)$-th percentile of this normal distribution:

$$g = \text{InvNorm}(\text{area} = 1 - \alpha, \mu = 175, \sigma = \frac{25}{\sqrt{n}})$$

Filling in $\alpha = 0.01$ and $n = 10$ gives $g \approx 193.3914$.

Exercise 9.19: In Exercise 9.15, calculate the p-value if the observed statistic is

(a) $\bar{x} = 180$ (b) $\bar{x} = 190$ (c) $\bar{x} = 170$

Answer

Notice that the hypothesis test of Exercise 9.15 can be formulated as $H_0 : \mu = 175$ versus $H_1 : \mu > 175$. That means that we are considering a right-sided hypothesis test.

In a hypothesis testing, we assume H_0 is true unless the evidence of the sample gives us significant reason to reject it. Under assumption of H_0, the mean foam height $\overline{X}$ is normally distributed with $\mu = 175$ and $\sigma = \frac{25}{\sqrt{n}}$. In a right-sided test, that means that the p-value (the probability that the test statistic $\overline{X}$ takes an even more extreme value than the observed value $\bar{x}$) is computed as $P(\overline{X} > \bar{x})$:

$$p = \text{normalcdf}(\text{lower} = \bar{x}, \text{upper} = 10^{99}, \mu = 175, \sigma = \frac{25}{\sqrt{15}})$$

(a) Filling in $\bar{x} = 180$ gives $p \approx 0.2193$.

(b) Filling in $\bar{x} = 190$ gives $p \approx 0.0101$.

(c) Filling in $\bar{x} = 170$ gives $p \approx 0.7807$.

Exercise 9.31: A hypothesis will be used to test that a population mean equals 4 against the alternative that the population mean is more than 4 with known standard deviation σ. What is the critical value for the test statistic Z_0 for the following significance levels?

(a) 0.01 (b) 0.05 (c) 0.10

Answer

We consider a hypothesis test with $H_0 : \mu = 4$ versus $H_1 : \mu > 4$ with a known standard deviation σ. As we work with a right-sided hypothesis test, the critical Z_0 value is

found at the $(1-\alpha)$-th percentile of the standard normal distribution, where α is the significance level of the hypothesis test:

$$Z_0 = \text{InvNorm}(\text{area} = 1-\alpha, \mu = 0, \sigma = 1).$$

(a) Filling in $\alpha = 0.01$ gives $Z_0 = 2.3263$

(b) Filling in $\alpha = 0.05$ gives $Z_0 = 1.6449$

(c) Filling in $\alpha = 0.10$ gives $Z_0 = 1.2816$

Exercise 9.33: For the hypothesis test $H_0 : \mu = 6$ against $H_1 : \mu \neq 6$ and variance known, calculate the p-value for each of the following test statistics.

(a) $z_0 = 2.05$ (b) $z_0 = -1.84$ (c) $z_0 = 0.4$

Answer

We consider a hypothesis test with $H_0 : \mu = 6$ versus $H_1 : \mu \neq 6$ with a known standard deviation σ. That means that we are considering a two-sided hypothesis test.

In a hypothesis testing, we assume H_0 is true unless the evidence of the sample gives us significant reason to reject it. Under assumption of H_0, the p-value (the probability that $\overline{X}$ takes value that lies even further from $\mu_0 = 6$ than the observed value $\overline{x}$). If $\overline{x}$ is smaller than $\mu_0 = 6$, this p-value equals $2 \cdot P(\overline{X} < \overline{x} | \mu = 6)$, whereas the p-value is computed as $2 \cdot P(\overline{X} > \overline{x} | \mu = 6)$ if $\overline{x} > \mu$.

This time, we do not look directly at the sample mean $\overline{X}$, but the corresponding z-value (transformed value to the standard normal distribution). The p-value can then be computed as:

$$p = \begin{cases} 2 \cdot P(\overline{Z} > z_0) = \text{normalcdf}(\text{lower} = z_0, \text{upper} = 10^{99}, \mu = 0, \sigma = 1), & \text{if } z \geq 0 \\ 2 \cdot P(\overline{Z} < z_0) = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = z_0, \mu = 0, \sigma = 1), & \text{otherwise.} \end{cases}$$

(a) Filling in $z_0 = 2.05$ gives $p = P(\overline{Z} > 2.05) = \text{normalcdf}(\text{lower} = 2.05, \text{upper} = 10^{99}, \mu = 0, \sigma = 1) \approx 0.0404$.

(b) Filling in $z_0 = -1.84$ gives $p = P(\overline{Z} < -1.84) = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = -1.84, \mu = 0, \sigma = 1) \approx 0.0658$.

(c) Filling in $z_0 = 0.4$ gives $p = P(\overline{Z} > 0.4) = \text{normalcdf}(\text{lower} = 0.4, \text{upper} = 10^{99}, \mu = 0, \sigma = 1) \approx 0.6892$.